

Probability Foundation for Machine Learning

Probability is the mathematical framework used to measure uncertainty. Since machine learning models make predictions on uncertain and incomplete data, probability forms the backbone of modern ML. ([GeeksforGeeks](#))

1. Why Probability in Machine Learning?

Imagine an email arrives.

Instead of saying:

"This email is definitely spam."

A machine learning model says:

"There is a 92% probability that this email is spam."

This ability to reason under uncertainty comes from probability theory. ([GeeksforGeeks](#))

Applications:

- Spam detection
- Medical diagnosis
- Weather prediction
- Stock forecasting
- Recommendation systems
- Autonomous vehicles

2. Basic Terminology

Experiment

An action whose outcome is uncertain.

Example:

- Tossing a coin
- Rolling a die
- Predicting whether a student passes an exam

Sample Space (S)

The set of all possible outcomes.

Example 1: Coin Toss

$S = \{H, T\}$ Example 2: Die Roll

$S = \{1, 2, 3, 4, 5, 6\}$ (CMU School of Computer Science)

Event

A subset of the sample space.

Example:

Rolling an even number.

$A = \{2, 4, 6\}$

3. Probability of an Event

For equally likely outcomes,

$$P(A) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Outcomes}}$$
Example

Probability of getting 4 on a die:

$$P(4) = \frac{1}{6}$$

Probability Range

$$0 \leq P(A) \leq 1$$
Meaning

Probability	Interpretation
0	Impossible
0.5	50% chance
1	Certain

([GeeksforGeeks](#))

4. Conditional Probability

Conditional probability means:

Probability of an event occurring given that another event has already occurred.

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
([CMU School of Computer Science](#))

Example

A class has:

- 60 students
- 30 are girls
- 15 girls know Python

Find:

$P(\text{Python} | \text{Girl}) = \frac{15}{30} = 0.5 = 0.5$ So there is a **50% chance that a student knows Python if we already know the student is a girl**.

5. Joint Probability

Probability that two events happen together.

Notation:

$P(A \cap B)$ ### Example

Probability that a student:

- Knows Python
- Knows Java

simultaneously.

6. Independent Events

Two events are independent if one does not affect the other.

Formula:

$P(A \cap B) = P(A)P(B)$ (CMU School of Computer Science)

Example

Event A:

Getting Head in a coin toss.

Event B:

Getting 6 on a die.

These events are independent.

$$P(A) = \frac{1}{2} \quad P(B) = \frac{1}{6} \quad P(A \cap B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

7. Bayes' Theorem

One of the most important concepts in Machine Learning.

Formula:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

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Intuition

Bayes theorem updates our belief when new evidence arrives.

Example

Medical Diagnosis

Before test:

- Disease probability = 1%

After positive test:

- Probability increases significantly.

This "belief updating" is Bayes theorem.

Used in:

- Naive Bayes Classifier
- Spam Filtering
- Medical Diagnosis
- Fraud Detection

8. Random Variable

A variable whose value depends on a random experiment.

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Example

Let X = Number obtained on a die.

Possible values:

$X = \{1, 2, 3, 4, 5, 6\}$

Types

Discrete Random Variable

Countable values.

Examples:

- Number of students absent
- Number of heads in coin tosses

Continuous Random Variable

Infinite possible values.

Examples:

- Height
- Weight
- Temperature

([CMU School of Computer Science](#))

9. Expected Value (Mean)

The long-term average outcome.

Formula:

$$E(X) = \sum xP(x)$$

Example

Fair Die

$$E(X) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \dots + 6\left(\frac{1}{6}\right) \\ = \frac{1+2+\dots+6}{6} = \frac{21}{6} = 3.5$$

Even though 3.5 never appears on a die, it is the average outcome over many rolls.

10. Probability Distributions

A probability distribution tells how probabilities are spread among possible values.

([SVGoudar](#))

Bernoulli Distribution

Only two outcomes.

Example:

- Pass/Fail
- Yes/No
- Spam/Not Spam

Binomial Distribution

Repeated Bernoulli trials.

Example:

Number of heads in 10 coin tosses.

Normal Distribution

Bell-shaped curve.

Many real-world measurements follow it:

- Height
- Weight
- Exam marks

Probability in Machine Learning: A Story

Imagine you are a teacher predicting whether a student will pass.

You observe:

- Attendance = 90%
- Assignments completed = Yes
- Internal marks = Good

From previous years:

- 95 out of 100 students with similar profiles passed.

So instead of saying:

"This student will definitely pass"

you say:

$P(\text{Pass}) = 0.95$ That is exactly how machine learning works.

The model learns probabilities from historical data and uses them to make predictions for new cases.

Key Concepts to Remember

1. Probability measures uncertainty.
2. Sample Space = all possible outcomes.
3. Event = subset of outcomes.
4. Conditional Probability:

$P(A|B)$ 5. Independence:

$P(A \cap B) = P(A)P(B)$ 6. Bayes Theorem updates beliefs.

6. Random Variables represent uncertain quantities.
6. Expected Value gives average behavior.
6. Probability Distributions model real-world data.
6. Machine Learning is fundamentally built on probability theory. ([GeeksforGeeks](#))

One-line exam definition

Probability is the branch of mathematics that quantifies uncertainty and measures the likelihood of events occurring, forming the foundation of machine learning models and decision-making under uncertainty. ([GeeksforGeeks](#))